

Climate and revenue considerations for the International Maritime Organization Net-Zero Framework

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INTRODUCTION

International shipping is a cornerstone of global trade, facilitating over 80% of global trade by volume.¹ It is also a significant contributor to greenhouse gas (GHG) emissions: in 2023, the sector emitted 785 million tonnes (Mt) of carbon dioxide equivalent (CO₂e) from fuel combustion, accounting for about 1.5% of anthropogenic GHG emissions.² The sector's contribution to global warming is even greater when accounting for black carbon emissions, a short-lived but strong climate forcer, and upstream emissions from fuel production and distribution.³

In 2023, the International Maritime Organization (IMO) revised its 2018 GHG strategy to achieve net-zero GHG emissions from international shipping by or around 2050. The strategy also mandated that the IMO develop mid-term measures, which include a technical element—a fuel standard regulating the life-cycle GHG intensity of marine fuels—and an economic element that requires ships to pay for their GHG emissions. The IMO approved the Net-Zero Framework in April 2025,⁴ but a vote to formally adopt the framework slated for October 2025 has been delayed for at least a year.⁵ Once implemented, all ships with a gross tonnage (GT) of 5,000 and above engaging in international voyages will be required to comply with this framework.

1 United Nations Conference on Trade and Development, *Review of Maritime Transport 2023*, 2023, https://unctad.org/system/files/official-document/rmt2023_en.pdf.

2 Xiaoli Mao et al., *Greenhouse Gas Emissions and Air Pollution from Global Shipping, 2016–2023* (International Council on Clean Transportation, 2025), <https://theicct.org/publication/greenhouse-gas-emissions-and-air-pollution-from-global-shipping-2016-2023-apr25/>.

3 Mao et al., *Greenhouse Gas Emissions*; Hae Jeong Cho et al., *Vision 2050: Fuel Standards to Align International Shipping with the Paris Agreement* (International Council on Clean Transportation, 2025), <https://theicct.org/publication/vision-2050-fuel-standards-to-align-international-shipping-with-the-paris-agreement-mar25/>.

4 International Maritime Organization, “IMO Approves Net-Zero Regulations for Global Shipping,” press release, April 11, 2025, <https://www.imo.org/en/mediacentre/pressbriefings/pages/imo-approves-netzero-regulations.aspx>.

5 International Maritime Organization, “IMO Net-Zero Shipping Talks to Resume in 2026,” press release, October 17, 2025, <https://www.imo.org/en/mediacentre/pressbriefings/pages/imo-net-zero-shipping-talks-to-resume-in-2026.aspx>.

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Core to the framework are two tiers of GHG fuel intensity (GFI) targets that become progressively more stringent over time. However, many elements of the framework remain undetermined, including GFI targets beyond 2035, noncompliance penalties beyond 2030, details pertaining to incentives for using zero- and near-zero-GHG fuels, guidelines for calculating the GFI of different fuels, and procedures for dividing and disbursing collected revenues.

This brief first provides an overview of the IMO Net-Zero Framework, with a focus on two key elements that, if addressed, could improve the framework's ability to reduce real-world GHG emissions and promote climate justice: the use of biofuels and funding for shipping-related climate mitigation.⁶ We first explore the impact of using food- and feed-based biofuels to meet the GFI targets, focusing on how emissions associated with land-use change could undermine the stated purpose of the framework to reduce GHG emissions. We then estimate the potential revenues that could be generated from penalties collected from ships that do not achieve GFI compliance targets and explain how the use of food- and feed-based biofuels could impact the revenue generated under the framework. We also explore potential climate mitigation and adaptation activities in the shipping sector that could be funded with the revenue, while investigating the gap between what the Net-Zero Framework might raise and the amount of shipping-related climate finance that might be needed to support robust mitigation and adaptation activities.

BACKGROUND

IMO NET-ZERO FRAMEWORK

The Net-Zero Framework was expected to be formally adopted in October 2025 and go into effect on March 1, 2027, with compliance starting January 1, 2028. However, the compliance date may shift to January 1, 2029, because of the vote to delay the adoption of the framework. The IMO has identified 13 different implementation workstreams, indicating the wide scope of matters that are still to be resolved.

The framework establishes two tiers of GFI targets—a Base target and a Direct Compliance target—which apply to the average well-to-wake life-cycle GHG emissions per unit of energy used on board a ship in a given year, expressed in grams of CO₂e per megajoule (g CO₂e/MJ).⁷ Both targets are calculated by applying reduction factors to the baseline value of 93.3 g CO₂e/MJ, which was the average GFI of international shipping in 2008. These reduction factors increase annually, resulting in progressively more stringent targets for each tier over time. Table 1 shows the GFI reduction factors in the current framework.⁸

6 This analysis was conducted prior to the vote to delay adoption of the framework. Because it is unclear if the framework's requirements will change, we analyzed the framework's existing requirements.

7 The GHG intensity of a fuel is calculated using GHG emission factors; these emission factors will be developed and incorporated into the IMO's life-cycle assessment (LCA) guidelines. Carbon dioxide, methane, and nitrous oxide are included in the calculation of emission factors. Emission factors are reported as CO₂ equivalents based on the global warming potentials for a 100-year time horizon. The current LCA guidelines use global warming potentials from the IPCC Fifth Assessment Report, which is 28 for methane and 265 for nitrous oxide.

8 Given the decision to delay adoption for at least 1 year, there is uncertainty around how these factors may change. There are several options: leave the targets unchanged and implement as normal (requires the collection of 2028 data), leave the targets unchanged but begin implementation in 2029, or begin implementation in 2029 and update the targets, or some other approach.

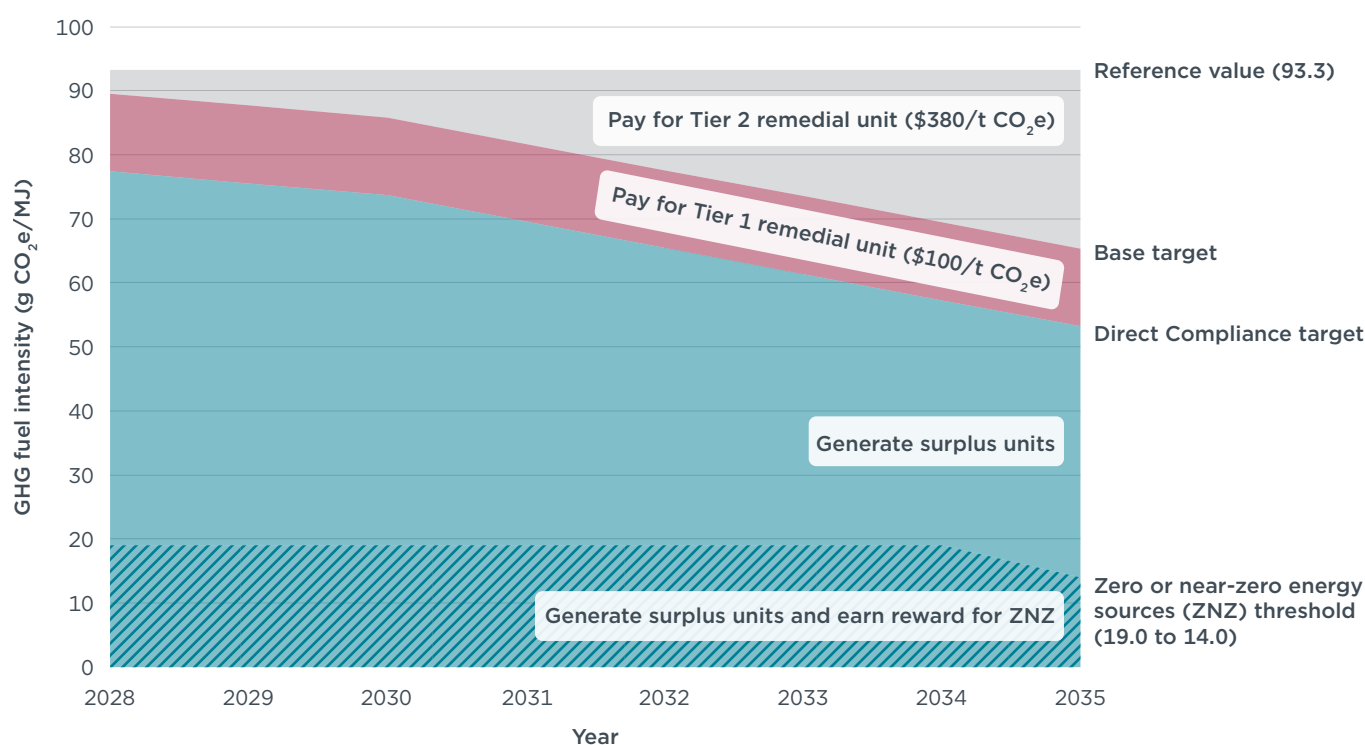
Table 1**Greenhouse gas fuel intensity (GFI) reduction factors and targets, 2028–2035**

Year	Base target		Direct Compliance target	
	GFI reduction factor (%)	GFI (g CO ₂ e/MJ)	GFI reduction factor (%)	GFI (g CO ₂ e/MJ)
2028	4.0%	89.57	17.0%	77.44
2029	6.0%	87.70	19.0%	75.57
2030	8.0%	85.84	21.0%	73.71
2031	12.4%	81.73	25.4%	69.60
2032	16.8%	77.63	29.8%	65.50
2033	21.2%	73.52	34.2%	61.39
2034	25.6%	69.42	38.6%	57.29
2035	30.0%	65.31	43.0%	53.18

The GFI targets for years beyond 2035 have not yet been defined by the IMO, except for the 2040 Base target, which is set at 65% below the 2008 value. The annual targets for the years 2036 to 2040 will be determined by January 1, 2032, and there is currently no deadline for setting the targets beyond 2040.

The two-tiered targets are tied to a complex penalty and reward mechanism (Figure 1). If a ship's GFI exceeds the less stringent Base target, the shipowner must pay a higher noncompliance penalty by purchasing Tier 2 remedial units (RUs); alternatively, they may buy credits or use banked credits earned from the ships that have over-complied with the Direct Compliance target (surplus units). If the GFI falls between the Base and Direct Compliance targets, the ship is subject to a lower noncompliance penalty (Tier 1 RUs). Shipowners are not permitted to avoid the Tier 1 penalty by buying or using surplus units. The prices of Tier 2 and Tier 1 RUs are \$380 and \$100 per tonne of CO₂e, respectively, through 2030. The mechanism to set RU prices starting in 2031 will be decided by the IMO after implementation.

Figure 1
Structure of the Net-Zero Framework, 2028–2035



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The IMO Net-Zero Fund revenues from these penalty fees will be used to reward the use of zero or near-zero GHG emission technologies, fuels, or energy sources (ZNZs), as well as to support climate mitigation, adaptation, and resilience activities relating to the energy transition in shipping. However, the IMO has not yet defined the reward amount for ZNZs. To qualify, ships will need to use low-emission fuels or other energy sources, such as wind-assisted propulsion, to achieve GFIs not exceeding 19 g CO₂e/MJ through 2034 and 14 g CO₂e/MJ from 2035 onward.

In addition to covering the administration costs of the Fund itself, the Fund will also focus on supporting developing countries, especially least developed countries (LDCs) and small island developing states (SIDS). Exactly how the Fund will distribute the revenues across these purposes is undecided.

BLENDING CROP-BASED BIOFUELS FOR COMPLIANCE

The current Net-Zero Framework is technology-neutral and does not exclude any fuel type; instead, fuels generate credits or deficits solely based on their direct supply chain life-cycle emissions. This means that ships will likely use fuels that offer the greatest direct GHG savings per dollar for compliance. One of the cheapest ways for shipowners to reduce GFI in the near term will be blending biofuels.

Shipowners looking to reduce emissions from their ships may choose to use drop-in biofuels such as hydroprocessed vegetable oils, also referred to as renewable diesel (RD). These biofuels can be produced using feedstocks such as waste oils and fats or virgin vegetable oils such as from soy or palm. Blending with these types of biofuels can be an attractive option because they are commercially available and relatively

cheap compared with nascent advanced fuels such as e-fuels and cellulosic biofuels.⁹ These fuels are also compatible with existing engine and fueling infrastructure and, depending on the fuel and the life-cycle assessment (LCA) approach used, could have lower GFIs than conventional fuels.¹⁰ For example, advanced fuels made from cellulosic wastes and residues could theoretically be produced at larger capacities without feedstock constraints, and could provide deeper decarbonization.¹¹ However, because their production requires significant upfront investment, the levelized cost of advanced fuels will not be competitive with hydroprocessed vegetable oil within the IMO framework.¹²

The GHG abatement costs of blending biofuels made from virgin vegetable or waste oils could be cost-effective relative to the Base target compliance cost of \$380/t CO₂e.¹³ However, waste oils and fats are scarce and are increasingly being used in other sectors for fuel production, including in sectors that have a higher willingness to pay than shipping, such as road transport and aviation.¹⁴ According to a previous ICCT analysis, if all available waste cooking oil used for biofuel production today were diverted for use in the shipping sector, the fuel produced would at most cover around 5% of the annual energy demand of ships subject to the Net-Zero Framework.¹⁵ Further, the framework could create demand for vegetable oil that is 3.5 times the amount currently used in the transport sector (40 billion liters in 2024), reaching a projected 139 billion liters by 2035.

LAND-USE CHANGE EMISSIONS FROM BIOFUELS

Food- and feed-based biofuels, such as those from palm and soy oil, are associated with high indirect land-use change (ILUC) emissions.¹⁶ ILUC occurs when farmland used for growing food is instead used for biofuel production.¹⁷ As a result, more land is cleared elsewhere for food production, causing the carbon stored in plants and soil to be released into the air, increasing the life-cycle GHG emissions from these fuels. This type of land-use change is different from direct land-use change (DLUC) emissions, which are associated with carbon stock changes directly attributable to the use or management of land to grow biofuel feedstocks.

- 9 Dan Rutherford, et al., *Feasibility Study of Future Energy Options for Great Lakes Shipping*, (International Council on Clean Transportation, 2024), <https://theicct.org/publication/feasibility-study-of-future-energy-options-for-great-lakes-shipping-march24/>.
- 10 Cato Sandford and Chris Malins, *Full Steam Ahead? Environmental Impacts of Expanding the Supply of Maritime Biofuels for the International Maritime Organisation Targets* (Cerulogy, February 2025), <https://www.transportenvironment.org/uploads/files/Cerulogy-Full-steam-ahead-February-2025.pdf>.
- 11 Jane O'Malley and Chelsea Baldino, *Availability of Biomass Feedstocks in the European Union to Meet the 2035 ReFuelEU Aviation SAF Target* (International Council on Clean Transportation, 2024), <https://theicct.org/publication/low-risk-biomass-feedstocks-eu-refueleu-aug24/>.
- 12 Andy Navarrete et al., *Industry Perspectives on Advanced Sustainable Aviation Fuel: What Barriers Remain for These Technologies to Scale?* (International Council on Clean Transportation, 2025) <https://theicct.org/publication/saf-what-barriers-remain-for-these-technologies-to-scale-jul25/>.
- 13 Transport & Environment, *IMO Net-Zero Framework: Assessing the Impact of the IMO's Draft Net-Zero Framework*, April 2025, <https://www.transportenvironment.org/uploads/files/Impact-of-the-IMOs-draft-Net-Zero-Framework-April-2025.pdf>; Gonca Seber and Nikita Pavlenko, *Estimating the Maximum Potential Vegetable Oil Demand for International Shipping under the International Maritime Organization's Net-Zero Framework* (International Council on Clean Transportation, 2025), <https://theicct.org/publication/estimating-the-maximum-potential-vegetable-oil-demand-for-international-shipping-under-the-imo-net-zero-framework-sept25/>.
- 14 Yuanrong Zhou et al., *The Potential of Liquid Biofuels in Reducing Ship Emissions* (International Council on Clean Transportation, 2020), <https://theicct.org/publication/the-potential-of-liquid-biofuels-in-reducing-ship-emissions/>.
- 15 Seber and Pavlenko, *Estimating the Maximum Potential*.
- 16 Hugo Valin et al., *The Land Use Change Impact of Biofuels Consumed in the EU: Quantification of Area and Greenhouse Gas Impacts* (BIENL13120) (European Commission, 2015), https://pure.iiasa.ac.at/id/eprint/12310/1/Final%20Report_GLOBIOM_publication.pdf.
- 17 Sammy El Takriti, et al., *Understanding Options for ILUC Mitigation* (International Council on Clean Transportation, 2016), https://theicct.org/wp-content/uploads/2021/06/ILUC-Mitigation-Options_ICCT_nov2016_0.pdf.

The IMO LCA guidelines account for DLUC emissions within the well-to-tank emission calculations, in accordance with Intergovernmental Panel on Climate Change guidelines.¹⁸ In addition, “carbon stock-direct land use change” is listed as a sustainability theme, along with metrics and indicators designed to restrict DLUC emissions from land conversion.¹⁹ While these criteria are important for preventing GHG emissions associated with the supply chain of a particular fuel, they do not account for indirect market effects such as ILUC emissions.

Because ILUC emissions cannot be directly measured, and are thus hard to quantify, economic models are used to estimate ILUC by evaluating how markets respond to changes in biofuel demand.²⁰ Despite quantification challenges, there is consensus that ILUC emissions could greatly impact the carbon intensity of biofuels.²¹ In some cases, such as fuels made from virgin palm oil, the scale of ILUC emissions could undermine the intended GHG savings of using biofuels entirely.²² Thus, to realize actual climate benefits from biofuels, it is important that ILUC effects are considered in fuels policies. For example, California’s Low Carbon Fuel Standard and the International Civil Aviation Organization’s Carbon Offsetting and Reduction Scheme (CORSIA) include ILUC emission factors estimated using various economic models, in addition to direct supply chain emissions from fuels.²³ Including this type of accounting could help limit the contribution of food- and feed-based biofuels that incur high ILUC emissions that are used for compliance under the Net-Zero Framework.

It is unclear how the Net-Zero Framework will account for ILUC emissions. The LCA guidelines list ILUC as a sustainability criterion and briefly describe a qualitative risk-based approach, referring to high-level definitions of low and high ILUC risk. Although ILUC emissions could erode or even eliminate potential GHG savings attributed to biofuels, the IMO’s current LCA guidelines do not assign ILUC emission factors to biofuels, and the framework does not limit the contribution of high-ILUC risk biofuels to achieving the GFI targets, thereby incentivizing the use of these biofuels for compliance.²⁴

CLIMATE FINANCE OVERVIEW

The IMO has stated that it will use revenues from the Net-Zero Fund for climate finance. To provide context for understanding and categorizing climate finance specific to international shipping, this section describes climate finance in general.

The United Nations Framework Convention on Climate Change (UNFCCC) defines climate finance as the mobilization of financial resources that:

18 International Maritime Organization, *Resolution MEPC.391(81) 2024 Guidelines on Life Cycle GHG Intensity of Marine Fuels*, March 22, 2024, <https://wwwcdn.imo.org/localresources/en/OurWork/Environment/Documents/annex/MEPC%2081/Annex%2010.pdf>.

19 For example, feedstocks for sustainable marine fuels do not include biomass obtained from high-carbon stock land. The production of these feedstocks does not occur in lands converted from primary forest, forestland, grassland or legally protected areas, taking January 1, 2008, as the cut-off date.

20 U.S. Environmental Protection Agency, *Model Comparison Exercise: Technical Document* (EPA-420-R-23-017), June 2023, <https://nepis.epa.gov/Exe/ZyPDF.cgi?Dockkey=P1017P9B.pdf>.

21 National Academies of Sciences, Engineering, and Medicine, *Current Methods for Life-Cycle Analyses of Low-Carbon Transportation Fuels in the United States*, 2022, <https://nap.nationalacademies.org/catalog/26402/current-methods-for-life-cycle-analyses-of-low-carbon-transportation-fuels-in-the-united-states>.

22 Yuanrong Zhou et al., *The Potential of Liquid Biofuels*.

23 “LCFS Land Use Change Assessment,” California Air Resources Board, accessed February 9, 2026, <https://ww2.arb.ca.gov/resources/documents/lcfs-land-use-change-assessment>; International Civil Aviation Organization, *CORSIA Supporting Document: CORSIA Eligible Fuels – Life Cycle Assessment Methodology Version 7*, June 2025, https://www.icao.int/sites/default/files/environmental-protection/CORSIA/Documents/CORSIA%20Eligible%20Fuels/CORSIA_Supporting_Document_CORSIA-Eligible-Fuels_LCA_Methodology_V7.pdf.

24 International Maritime Organization, *Resolution MEPC.391(81)*; WWF et al., *Developing a Quantitative Risk-Based Approach to Manage Indirect Land Use Change Emissions under the IMO Net-Zero Framework* (ISWG-GHG 20/3/8), September 5, 2025, <https://cleanshipping.org/publication/iswg-ghg-20-3-8-developing-a-quantitative-risk-based-approach-to-manage-indirect-land-use-change-emissions-under-the-imo-net-zero-framework/>.

*aims at reducing emissions and enhancing sinks of greenhouse gases; aims at reducing vulnerability, increasing adaptive capacity, and mainstreaming and increasing resilience of human and ecological systems to negative climate impacts; and includes financing for actions identified in a country's nationally determined contribution, adaptation communication, national adaptation plan, long-term low-emission development strategy or other national plan for implementing and achieving the goals of the Paris Agreement and the objective of the Convention.*²⁵

Climate finance flows are often disaggregated by financial institutions by area of support (e.g., mitigation, adaptation, or both), funding source (e.g., public or private, domestic or international), financial instrument (e.g., grant, loan, or equity), and destination (e.g., by region or income group).²⁶

The operational definitions of climate mitigation and adaptation finance can vary by financing and data-collecting institution.²⁷ The Climate Policy Initiative defines mitigation finance as directing resources “to activities contributing to reducing or avoiding GHG emissions, including gases regulated by the Montreal Protocol; or maintaining or enhancing GHG sinks and reservoirs.”²⁸ The initiative defines adaptation finance as directing resources “to activities aimed reducing the vulnerability of human or natural systems to the impacts of climate change and climate-related risks, by maintaining or increasing adaptive capacity and resilience.”²⁹

Dual benefits finance, also referred to as cross-cutting finance in the UNFCCC, refers to directing resources to activities that address both mitigation and adaptation.³⁰ An afforestation project that helps both to sequester carbon and to reduce soil erosion would be an example of dual benefits financing.³¹

Global climate finance more than doubled from \$812 billion in 2018 to \$1.9 trillion in 2023, although the distribution was uneven across areas of support, funding sources, and funding destinations.³² For example, mitigation finance accounted for 94% of the total flows in 2023. The private sector contributed to two-thirds of the flows, growing faster than public-sector contributions, and 87% of the flows were raised and spent domestically. The slower growth of public-sector contributions could be a concern for adaptation finance as well as climate finance provided to LDCs and SIDS, both of which rely heavily on public sector contributions. Despite rapid growth, global climate finance mobilization remains well below the levels required to keep global warming below 1.5 °C, estimated to be \$6.7–\$10 trillion annually through 2050.³³

An emerging area of support is addressing loss and damage, which generally refer to impacts of climate change remaining after mitigation and adaptation measures have been implemented.³⁴ Examples include damage to infrastructure or loss of life or

25 United Nations Framework Convention on Climate Change, *Sixth Biennial Assessment and Overview of Climate Finance Flows*, 2024, https://unfccc.int/sites/default/files/resource/UNFCCC_BA6_Report_Web_Apr2025.pdf.

26 Climate Policy Initiative, *Global Landscape of Climate Finance 2023 Methodology*, November 2023, <https://www.climatepolicyinitiative.org/wp-content/uploads/2023/11/GLCF-2023-Methodology.pdf>.

27 United Nations Framework Convention on Climate Change, *Sixth Biennial Assessment*.

28 Climate Policy Initiative, *Global Landscape of Climate Finance 2023*.

29 Climate Policy Initiative, *Global Landscape of Climate Finance 2023*.

30 United Nations Framework Convention on Climate Change, *Sixth Biennial Assessment*.

31 Climate Policy Initiative, *Global Landscape of Climate Finance 2023*.

32 Baysa Naran et al., *Global Landscape of Climate Finance 2025* (Climate Policy Initiative, June 2025), <https://www.climatepolicyinitiative.org/publication/global-landscape-of-climate-finance-2025/>.

33 Baysa Naran et al., *Global Landscape of Climate Finance 2024: Insights for COP 29* (Climate Policy Initiative, October 2024), <https://www.climatepolicyinitiative.org/publication/global-landscape-of-climate-finance-2024/>.

34 Reinhard Mechler et al. (Eds.), *Loss and Damage from Climate Change: Concepts, Methods and Policy Options*, Climate Risk Management, Policy and Governance (Springer International Publishing, 2019), <https://doi.org/10.1007/978-3-319-72026-5>.

culture.³⁵ However, addressing loss and damage is not listed as a potential use of the revenues under the IMO Net-Zero Framework. As such, the following section on climate finance for international shipping focuses on mitigation and adaptation activities.

CLIMATE FINANCE FOR INTERNATIONAL SHIPPING

The Net-Zero Framework states that the revenues under the Net-Zero Fund will be used to reward the use of ZNZs and to facilitate “environmental and climate protection, adaptation and resilience building within the boundaries of the energy transition in shipping.”³⁶ The Framework also specifies a particular focus on LDCs and SIDS, allocating revenue for the development and deployment of ZNZs, as well as support for necessary port infrastructure; enabling a just transition for the maritime workforce and addressing disproportionately negative impacts on States, including on food security; and supporting the development and implementation of National Action Plans, including fleet renewal and upgrade.

Mitigating GHG emissions from shipping will require a fundamental transformation of energy sources, propulsion systems, and port and fueling infrastructure. Climate finance can play a catalytic role in de-risking these investments, supporting early deployment of emerging technologies, and reducing the cost gap between conventional fuels and ZNZs. Examples of mitigation activities within the shipping sector are shown in Table 2.

Table 2
Mitigation activities related to shipping sector

Mitigation activity	Example
Fuel production and infrastructure deployment	Investing in the production, distribution, and bunkering infrastructure for alternative fuels, renewable ammonia, methanol, and hydrogen. ^a
Fleet decarbonization	Retrofitting existing vessels with dual-fuel or zero-emission propulsion systems; supporting the construction of zero-emission capable ships. ^b
Propulsion and energy efficiency improvements	Financing the uptake of technologies such as wind-assisted propulsion and air lubrication systems to reduce fuel consumption and emissions. ^c
Port electrification and renewable power integration	Supporting onshore power supply systems and renewable energy generation at ports to reduce emissions from ships at berth. ^d

^a RTI International, “Exploring Green Ammonia as a Sustainable Maritime Fuel,” November 14, 2023, <https://www.rti.org/impact/green-ammonia-as-sustainable-maritime-fuel>.
^b Lisa Baertlein, “Container Shippers Hedging Green Transition with Dual-Fuel Vessel Orders,” *Reuters*, November 21, 2024, <https://www.reuters.com/sustainability/climate-energy/container-shippers-hedging-green-transition-with-dual-fuel-vessel-orders-2024-11-21/>.
^c International Maritime Organization, *2021 Guidance on Treatment of Innovative Energy Technologies for Calculation and Verification of the Attained EEDI and EEXI* (MEPC.1/Circ.896), December 14, 2021, <https://wwwcdn.imo.org/localresources/en/OurWork/Environment/Documents/Air%20pollution/MEPC.1-Circ.896.pdf>.
^d California Air Resources Board, *Interim Evaluation Report: Control Measure for Ocean-Going Vessels At Berth*, December 1, 2022, https://ww2.arb.ca.gov/sites/default/files/2022-12/At%20Berth%20Interim%20Evaluation%20Report_Final_Remediated.pdf.

35 In 2013, the UNFCCC established the Warsaw International Mechanism to address loss and damage in developing countries that are particularly vulnerable to the adverse effects of climate change. Almost a decade later at the 2022 United Nations climate summit (COP27), countries agreed to create the Fund for Responding to Loss and Damage, to which \$768 million had been pledged as of March 2025. See: United Nations Framework Convention on Climate Change, *Report of the Conference of the Parties on its Nineteenth Session, Held in Warsaw from 11 to 23 November 2013. Addendum. Part Two: Action Taken by the Conference of the Parties at its Nineteenth Session* (FCCC/CP/2013/10/Add.3), January 31, 2014, <https://unfccc.int/documents/8105>; “Funding,” Fund for Responding to Loss and Damage, accessed June 30, 2025, <https://www.fird.org/pledges>.
36 International Maritime Organization, *Circular Letter No.5005* (2025), [https://wwwcdn.imo.org/localresources/en/MediaCentre/HotTopics/Documents/Circular%20Letter%20No.5005%20-%20Draft%20Revised%20Marpol%20Annex%20Vi%20\(Secretariat\).pdf](https://wwwcdn.imo.org/localresources/en/MediaCentre/HotTopics/Documents/Circular%20Letter%20No.5005%20-%20Draft%20Revised%20Marpol%20Annex%20Vi%20(Secretariat).pdf).

A 2020 brief estimated that the scale of investments needed to decarbonize shipping by 2050 would be \$1.4–\$1.9 trillion between 2030 and 2050, or \$70–\$95 billion annually, assuming ammonia is used as the primary alternative fuel.³⁷ The higher estimate was projected based on the assumption that ammonia was produced using renewable hydrogen. About 87% of the investments would be used for hydrogen production, ammonia synthesis, and land-based storage and bunkering infrastructure. The rest would be used for funding ship-side purchases, retrofits, and efficiency improvements. The estimate of \$95 billion per year did not include investments needed for renewable electricity production.

At the same time, shipping is vulnerable to the impacts of climate change and requires adaptation. For example, ports and waterways can be affected by sea-level rise or droughts, and ships and infrastructure are affected by severe storms. As a result, there can be damage to infrastructure, delays and disruptions in shipping and port operation, and loss of life and vessels.³⁸ Because they depend on shipping for trade, food, energy security, and tourism, SIDS and other island nations are particularly vulnerable to these impacts.³⁹ Adaptation finance from the Net-Zero Fund could help the maritime sector prepare for and respond to growing climate risks, particularly for SIDS and LDCs. Adaptation examples within the shipping sector are shown in Table 3.

Table 3
Adaptation activities related to shipping sector

Adaptation activity	Example
Climate-resilient port infrastructure	Investing in elevated docks, storm surge barriers, and flood protection systems. ^a
Technological resilience	Integrating dynamic routing systems that optimize voyage paths by analyzing real-time weather forecasts, ocean currents, and port congestion data. ^b
Climate risk assessments and long-term planning	Conducting vulnerability assessments of maritime infrastructure to inform long-term adaptation strategies and investment decisions, which can inform National Action Plans. ^c

^a George Van Houtven et al., *Act Now or Pay Later: The Costs of Climate Inaction for Ports and Shipping* (RTI International, March 14, 2022), <https://www.edf.org/sites/default/files/press-releases/RTI-EDF%20Act%20Now%20or%20Pay%20Later%20Climate%20Impact%20Shipping.pdf>.

^b Mark Ching-Pong Poo and Zaili Yang, “Optimising the Resilience of Shipping Networks to Climate Vulnerability,” *Maritime Policy & Management* 51, no. 1 (2024): 15–34, <https://doi.org/10.1080/03088839.2022.2094488>.

^c Cristina Izaguirre et al., “Seaport Climate Change Impact Assessment Using a Multi-Level Methodology,” *Maritime Policy & Management* 47, no. 4 (2020): 544–557, <https://www.tandfonline.com/doi/full/10.1080/03088839.2020.1725673>.

METHODS

ENERGY DEMAND SUBJECT TO THE NET-ZERO FRAMEWORK

To estimate potential ILUC emissions and revenues generated by the Net-Zero Framework between 2028 and 2035, we first projected the amount of energy consumed by ships subject to the framework over the same period. We used the

37 Randall Krantz et al., *The Scale of Investment Needed to Decarbonize International Shipping* (Global Maritime Forum, January 20, 2020), <https://globalmaritimeforum.org/news/the-scale-of-investment-needed-to-decarbonize-international-shipping/>.

38 George Van Houtven et al., *Act Now or Pay Later: The Costs of Climate Inaction for Ports and Shipping* (RTI International, March 2022), <https://www.edf.org/sites/default/files/press-releases/RTI-EDF%20Act%20Now%20or%20Pay%20Later%20Climate%20Impact%20Shipping.pdf>.

39 Regina Asariotis, *Climate Change Impacts on Seaports: A Growing Threat to Sustainable Trade and Development* (UN Trade & Development, June 4, 2021), <https://unctad.org/isar/news/climate-change-impacts-seaports-growing-threat-sustainable-trade-and-development>.

energy demand projections from the ICCT's Polaris model version 1.3.⁴⁰ Polaris is a global maritime emissions projection model covering all ocean transport activity of Type 1 and Type 2 vessels as defined in the Fourth IMO GHG Study;⁴¹ the emissions of these vessels have been found to be a good approximation of international shipping emissions.⁴² Energy demand is one output of the model and takes into account existing IMO policies on energy efficiency, such as the Energy Efficiency Design Index and Energy Efficiency Existing Ship Index. We only considered the energy demand for ships expected to be subject to the Net-Zero Framework, namely, those that are 5,000 GT and above.

COMPLIANCE STRATEGY FOR NET-ZERO FRAMEWORK

In a study estimating demand for vegetable oil under the International Maritime Net-Zero Framework, Seber and Pavlenko found that the use of biofuels is cost-effective relative to the Tier 2 RU penalty of \$380/t CO₂e but more expensive than the Tier 1 penalty of \$100/t CO₂e.⁴³ Therefore, we assumed that shipowners would minimize compliance costs by meeting the Base targets to avoid the Tier 2 penalty and pay only the Tier 1 penalty. In line with Seber and Pavlenko, we also assumed that used cooking oil-RD would be blended first up to its supply limits before the other vegetable oil-RD is blended.

POTENTIAL ILUC EMISSIONS FROM NET-ZERO FRAMEWORK

To estimate the fuel carbon intensity including ILUC emissions until 2035, we used the potential vegetable oil demand for blending estimated in Seber and Pavlenko.⁴⁴ We assumed that this demand would be met by soy-RD, which we used as an example for ILUC estimations. For ILUC emissions, we used an illustrative value of 62 g CO₂e/MJ from a U.S. Environmental Protection Agency model comparison exercise.⁴⁵ Although this value may be perceived as high for the United States compared with other estimations,⁴⁶ it could also understate the ILUC emissions associated with feedstocks such as Indonesian palm oil or Brazilian soy oil, where ILUC emissions have been found in some studies to exceed 100 g CO₂e/MJ.⁴⁷

Because we assumed that the fuel blend would meet the Base targets, we calculated the final fuel carbon intensity by adding the ILUC emissions associated with soy-RD to the Base target GFI.

POTENTIAL REVENUES FROM THE NET-ZERO FRAMEWORK

To estimate the revenues that the IMO will collect for the Net-Zero Fund between 2028 and 2035, we assumed that the RU prices would remain constant through 2035, although the prices beyond 2031 have not been determined.

Between 2028 and 2035, the difference between Base and Direct Compliance targets remains constant at 12.13 g CO₂e/MJ, as shown in Table 1. Assuming that shipowners

40 Gabriel Hillman Alvarez et al., *Polaris v1.3 Documentation* (International Council on Clean Transportation, 2025), <https://theicct.github.io/polaris-doc/versions/v1.3/>.

41 Type 1 and Type 2 vessels are vessels registered in the Automatic Identification System; they can be identified by either their IMO number (Type 1) or Maritime Mobile Service Identity number (Type 2).

42 Jasper Faber et al., *Fourth IMO Greenhouse Gas Study 2020* (International Maritime Organization, 2020), <https://wwwcdn.imo.org/localresources/en/OurWork/Environment/Documents/Fourth%20IMO%20GHG%20Study%202020%20-%20Full%20report%20and%20annexes.pdf>; Cho et al., *Vision 2050*.

43 Seber and Pavlenko, *Estimating the Maximum*.

44 Seber and Pavlenko, *Estimating the Maximum*.

45 U.S. Environmental Protection Agency, *Model Comparison Exercise*.

46 U.S. Environmental Protection Agency, *Summary Table of Lifecycle Greenhouse Gas Emissions for Select Pathways*, accessed January 25, 2026, <https://www.epa.gov/fuels-registration-reporting-and-compliance-help/summary-table-lifecycle-greenhouse-gas-emissions>.

47 Valin et al., *Land Use Change Impact*.

would pay \$100/t CO₂e for this 12.13 g CO₂e/MJ difference, the total annual compliance costs to be paid to the Net-Zero Fund annually was calculated using Equation 1.⁴⁸

Compliance cost (\$) =
= \$100/t CO₂e
× Energy consumption from ships of 5,000 GT and above (MJ)
× 12.13 gCO₂e/MJ ÷ 10⁶ g CO₂e/t CO₂e

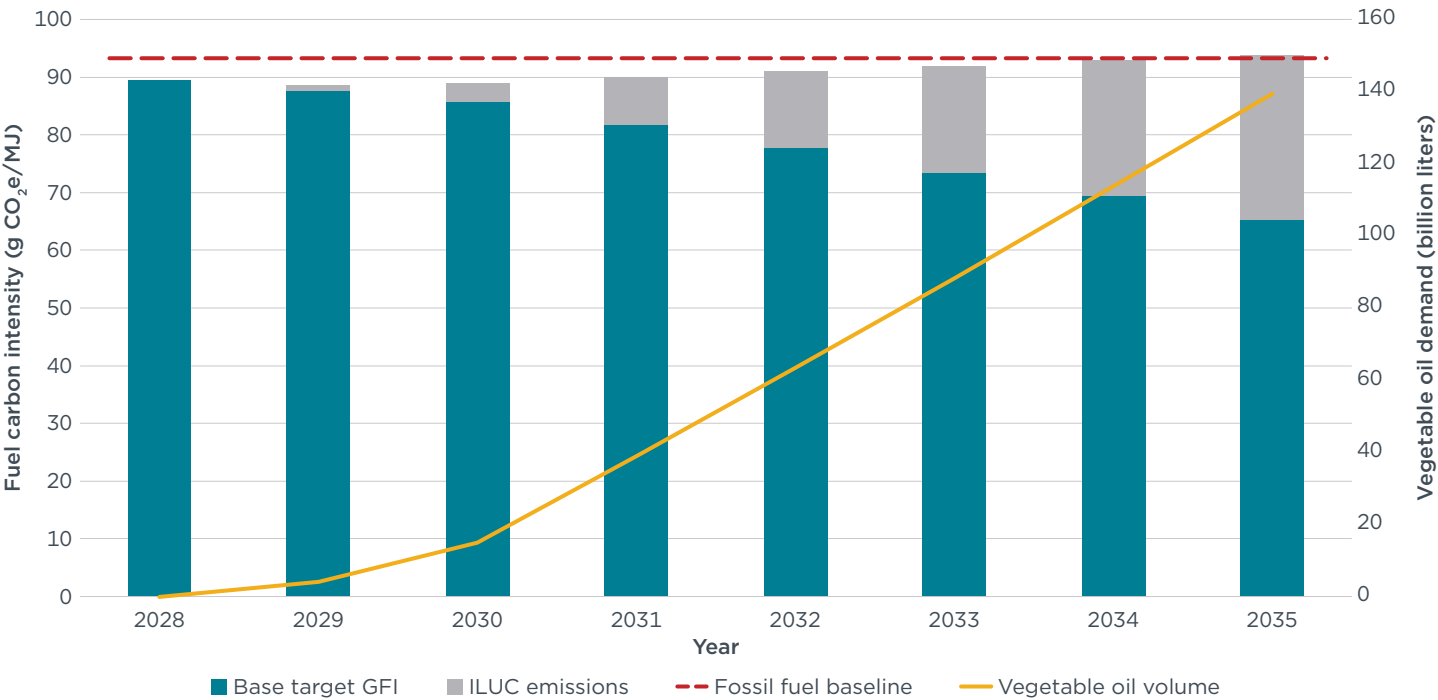
Equation 1

RESULTS AND DISCUSSION

ILUC IMPLICATIONS

Figure 2 illustrates the Base targets of the Net-Zero Framework and the estimated contributions from ILUC emissions, assuming soy-RD is blended to comply. The results indicate that, by 2035, the carbon intensity of the fuel mix containing vegetable oils could surpass the reference value used to calculate GFI targets—that is, the average GFI of international shipping in 2008. In our illustrative example, the contribution of ILUC emissions to the overall GFI start from 0.9 g CO₂e/MJ in 2029 and increase to 28.5 g CO₂e/MJ by 2035 (see Table A1 in the Appendix). The annual ILUC emissions would increase to approximately 285 Mt CO₂e by 2035, with aggregate ILUC emissions of around 950 Mt CO₂e between 2028 and 2035, although with a high degree of uncertainty (e.g., around which food and feed biofuel would be used for compliance). These ILUC emissions alone are comparable to the estimated annual well-to-wake GHG emissions from international shipping in 2024 (939 Mt CO₂e).⁴⁹

Figure 2
Virgin vegetable oil demand that may be induced by the Net-Zero Framework and the associated ILUC emissions



Note: These results are based on the assumption that soy-renewable diesel is used for blending, with an ILUC factor of 62 g CO₂e/MJ. The projection of vegetable oil demand was taken from Seber and Pavlenko, *Estimating the Maximum*.

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48 10⁶ is the conversion factor from grams to tonnes.

49 Cho et al., *Vision 2050*.

While the Net-Zero Framework sets substantial GHG reduction targets on paper, Figure 2 illustrates how loose eligibility criteria for biofuels threaten its goal of reducing GHG emissions in the maritime sector. In the absence of safeguards, the structure of the framework is likely to create a strong demand for virgin vegetable oils as shipping companies try to meet the growing compliance obligations with drop-in fuels. Accordingly, converting additional land for cultivation or shifting vegetable oil supply away from non-transport sectors could be necessary. Considering the amount of vegetable oils used for biofuels production today (40 billion liters), we estimate that the supply of virgin vegetable oils would be insufficient to meet the GFI targets beyond 2032, and later-year targets may be unachievable by these biofuels alone.

Moreover, blending biofuels to comply with the Base target will likely fail to deliver the intended emission reductions. This is because biofuels made from palm or soy have high ILUC emissions that can negate all emission savings from these biofuels.⁵⁰ At the same time, waste and residue-based biofuels that could be used for compliance are limited in supply.⁵¹ Protections against using biofuels that pose the risk of market-induced land-use impacts can therefore ensure genuine decarbonization and avoid the effects of unintended ILUC emissions.

REVENUE PROJECTIONS

We calculated that, from 2028 to 2035, owners of ships of 5,000 GT or more would pay approximately \$11.4 to \$12.1 billion annually to the IMO Net-Zero Fund (Table 4). In line with projected energy demand growth, the amount would increase annually by 0.8%–1.1%. For comparison, Smith et al. similarly estimated that the Net-Zero Framework would generate \$11–\$12 billion in revenue annually between 2028 and 2030,⁵² while Transport & Environment estimated annual revenues of approximately \$10 billion.⁵³

Table 4
Annual energy demand and revenue from ships subject to the Net-Zero Framework, 2028–2035

Year	Energy demand (EJ)	Revenue (\$ billion)
2028	9.4	11.4
2029	9.5	11.5
2030	9.6	11.6
2031	9.7	11.7
2032	9.7	11.8
2033	9.8	11.9
2034	9.9	12
2035	10	12.1

The estimated revenue is greater than those of climate funds under the UNFCCC: the Green Climate Fund pledged a total of \$10.6 billion for the period 2024–2027

50 Valin et al., *Land Use Change Impact*.
51 Seber and Pavlenko, *Estimating the Maximum*.
52 Tristan Smith et al., “An Overview of the Discussions from IMO’s 83rd Marine Environment Protection Committee” (UCL Shipping and Oceans Research Group, April 11, 2025), <https://www.ucl.ac.uk/bartlett/sites/bartlett/files/2025-05/An%20overview%20of%20the%20discussions%20from%20IMO%E2%80%99s%2083rd%20Marine%20Environment%20Protection%20Committee.pdf>.
53 Transport & Environment, *IMO Net-Zero Framework*.

(approximately \$2.7 billion per year on average)⁵⁴ and the Global Environment Facility pledged \$5.35 billion for the period 2022–2026 (approximately \$1.3 billion per year on average).⁵⁵ Yet, projected annual revenue of \$11–\$12 billion is not enough to cover costs related to the energy transition in shipping. Considering the previously cited estimate of \$95 billion annually to decarbonize shipping by 2050 using renewable ammonia, this revenue would cover just 12%–13% of the amount required, not including renewable electricity generation or adaptation costs.⁵⁶ The IMO could consider this substantial gap when setting RU prices for 2031 and beyond.

However, simply increasing the RU prices without safeguards against inexpensive biofuels could backfire. Shipowners are likely to choose to blend biofuels to avoid paying for any Tier 2 RUs at the price of \$380/t CO₂e. Increasing the Tier 2 RU price would therefore not raise any additional revenues. A higher Tier 1 RU price, on the other hand, could make blending low-cost biofuels to meet the Direct Compliance target a more attractive option, which could collapse revenue generation and leave even fewer resources in the Net-Zero Fund. For example, shipowners would likely not blend biofuels to avoid the current Tier 1 RU price of \$100/t CO₂e but would be much more likely to do so at an RU price of \$300/t CO₂e.

SUMMARY AND POLICY CONSIDERATIONS

This briefing explores two elements of the Net-Zero Framework that, if addressed, would improve the framework's ability to reduce real-world GHG emissions and promote climate justice: the use of biofuels for compliance and funding for shipping-related climate mitigation. Food and feed-based biofuels could be used to comply with the Net-Zero Framework. However, these biofuels are associated with high ILUC emissions that could undermine their emission benefits, requiring that proper safeguards be introduced. Using food- and feed-based biofuels could also impact the revenues generated under the framework, as they would become a more attractive option for shipowners than paying base target penalties. The projected revenues generated under the Net-Zero Framework are unlikely to cover the cost for an energy transition in shipping, as they are estimated to cover only a small fraction of the amount needed.

This analysis supports the following policy considerations.

Strong protections against using fuels that pose a high risk of ILUC impacts can ensure that the Net-Zero Framework catalyzes genuine decarbonization. Including ILUC emissions accounting in the LCA guidelines or limiting the contribution of food- and feed-based biofuels would help prevent adverse effects from these emissions.

Allowing the use of food and feed biofuels, in particular vegetable oils, could completely undermine the GHG emissions reduction intended by the Net-Zero Framework. As illustrated here, if blending with food- and feed-based biofuels is assumed to comply with the base GFI trajectory, the ILUC-adjusted GFI of the fuel blend could exceed the IMO baseline of 93.3 g CO₂e/MJ by 2035. Existing qualitative risk assessment framework proposals at the IMO fail to address how such high-risk fuels should be constrained within the Net-Zero Framework.⁵⁷ FuelEU Maritime classifies food and feed biofuels as equivalent to fossil fuels in emission calculations;

54 Green Climate Fund, *Status of Pledges and Contributions*, July 31, 2025, <https://www.greenclimate.fund/sites/default/files/document/2025-status-pledges-website-july-31.pdf>.

55 Deloitte and Touche, LLC, *Global Environment Facility Trust Fund (GEF): Independent Auditor's Report and Statements of Receipts, Disbursements and Fund Balance* (TF029840), June 30, 2024, https://fiftrustee.worldbank.org/content/dam/fif/funds/gef/FinancialStatements/GEF%20FS%20FY24%2006_2024.pdf.

56 Krantz et al., *The Scale of Investment*.

57 WWF et al., *Developing a Quantitative Risk-Based Approach*.

the IMO could similarly exclude or limit food and feed biofuels, especially virgin vegetable oils, in the Net-Zero Framework. Introducing ILUC emissions accounting as in CORSIA would also help limit the certification of food and feed biofuels.

Limiting food and feed biofuels would also facilitate the expansion of emerging advanced renewable fuel technologies, such as cellulosic biofuels or e-fuels. While these alternative renewable fuels cannot yet compete economically with established hydroprocessed vegetable oil technology, they will be essential for meeting the IMO's decarbonization goals as they could be produced at larger capacities without feedstock constraints while enabling meaningful decarbonization.

To ensure that the revenue from the Net-Zero Framework supports scalable solutions to shipping decarbonization, IMO member states could consider limiting financial incentives exclusively to advanced fuels.

We estimate that the current framework would generate an annual revenue of approximately \$11–\$12 billion, a substantial amount—though insufficient to cover the investments needed to decarbonize the sector, estimated to be as much as \$95 billion per year when assuming renewable ammonia as the primary alternative fuel. IMO member states could thus consider limiting financial incentives, such as the ability to generate surplus units and rewards for ZNZs, to only advanced, second-generation biofuels, such as those produced from cellulosic materials and renewable hydrogen and its derivatives. These fuels offer a scalable—although more expensive—solution to shipping decarbonization that can deliver higher GHG savings than existing biofuels.⁵⁸ Focusing financial incentives on these fuels would help lower their costs.

58 Nikita Pavlenko and Chelsea Baldino, “Getting Investors to Say Yes to Low-GHG Fuels for Aviation and Marine: Opportunities in Europe’s Clean Industrial Deal,” *International Council on Clean Transportation Staff Blog*, March 4, 2025, <https://theicct.org/getting-investors-to-say-yes-to-low-ghg-fuels-for-aviation-and-marine-mar25/>.

APPENDIX

Table A1

Indirect land-use change (ILUC) emission estimates

Base target GFI (g CO ₂ e/MJ)	ILUC emissions of fuel blend (g CO ₂ e/MJ)	ILUC-adjusted GFI (g CO ₂ e/MJ)	Shipping energy demand (EJ)	Cumulative ILUC emissions (Mt CO ₂ e)
89.57	0.00	89.57	9.4	0.0
87.70	0.89	88.59	9.5	8.5
85.84	3.21	89.05	9.6	30.8
81.73	8.28	90.01	9.7	80.0
77.63	13.34	90.97	9.7	129.9
73.52	18.41	91.93	9.8	180.9
69.42	23.48	92.90	9.9	233.0
65.31	28.55	93.86	10.0	285.5



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